A-A-1016B
April 28, 1987
SUPERSEDING
A-A-1016A
April 3, 1984

COMMERCIAL ITEM DESCRIPTION

DISK, ABRASIVE
(ALUMINUM OXIDE, ALUMINA ZIRCONIA AND CERAMIC ALUMINUM OXIDE)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification P-D-455.

This description covers industrial quality abrasive coated disks used in portable sanding machines for polishing, finishing, grinding and removal of rust and scale.

Salient characteristics:

DISK Outside Diameter Inches	DISK ANCHOR HOLE Diameter Inches
9-1/8, 7, 6, 5, 4	1/2, 5/8, 7/8

Tolerances. The outside diameter shall have a plus or minus 1/16 inch tolerance. The arbor hole shall have a plus 1/16 inch, minus 0 tolerance.

Backing Material. The coated abrasive disks shall have a backing of vulcanized fiber or a functionally similar synthetic. The backing shall have the strength and flexibility to grind surfaces at high speeds. The backing material shall be not less than 0.023 inch thick for grit sizes 80 and finer and not less than 0.028 inch thick for grit sizes 60 and coarser.

Adhesive. The adhesive shall be of good quality, nonhydroscopic, made from synthetic resins and resistant to severe heat and humidity conditions.

Abrasive. The abrasive shall be aluminum oxide, alumina zirconia or ceramic aluminum as specified. The abrasive shall be 95 percent free of extraneous material.

Grit sizes (Grades). The coated abrasive disks shall be in the following grit sizes as specified: Closed coat: 180, 150, 120, 100, 80, 60, 50, 36, 24, 16; Open coat: 50, 36, 24, 16.

FSC 5345

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Closed coat. The surface of the backing shall be completely and uniformly covered by the abrasive grain.

Open coat. The abrasive grain shall uniformly cover 50 to 70 percent of the backing.

Grading. The grading, control and tolerance of the abrasive shall be in accordance with ANSI B74.18, `Specifications for Grading of Abrasive Grain on Coated Abrasive Products'.

Slits. The slits in disks shall be radial incisions, equidistantly spaced and have no appreciable width.

Slots. Equidistantly spaced and radially arranged slots shall have a 1/16 to 1/8 inch width.

Breaking strength. The average breaking strength of the coated abrasive disks shall be not less than 350 pounds per inch of width in the machine direction and 180 pounds in the transverse fiber direction as determined by Method ASTM D828, `Tensile breaking strength of paper and paperboard, Test methods for', except that the jaws shall be five inches apart and five specimens each, either one-half inch or one inch wide, shall be taken longitudinally and transversely.

Marking. Coated abrasive disks shall be marked in a legible manner with the manufacturer's name or trademark, kind of abrasive, open coat designation if applicable and grit size.

Workmanship. Cracks, uneven distribution of grit, frayed edges, or other imperfections shall not be present to impair the serviceability of the abrasive disks.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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Sampling.

Lot. All disks of the same size and grade presented for inspection at one time shall be considered a lot for purposes of inspection.

Sampling for quality conformance examination. A random sample of disks shall be selected from each lot in accordance with Military Standard MIL-STD-105, 'SAMPLING PROCEDURES AND TABLES FOR INSPECTION BY ATTRIBUTES', special inspection level S-2, AQL 1.5 expressed in terms of percent defective.

Visual and dimensional inspection. The sample disks shall be visually and dimensionally examined to verify compliance with this specification.

Certification of grading and breaking strength. When specified in the contract or purchase order, a certificate of conformance may be furnished in lieu of actual testing on a lot by lot basis providing that:

(1) Objective evidence dated not more than two (2) years prior to the date of the start of the contract as described in the solicitation or date of the purchase order as applicable shall be available.
(2) Tests specified in the industry standards referenced herein have been performed by an independent test facility acceptable to the Government.
(3) That the product being offered is the same type and grade of abrasive and backing material and has been manufactured using the same material, processes and manufacturing techniques as the product tested.

Examination for preparation for delivery. A random sample of unit, intermediate and shipping containers (as applicable) shall be selected from each lot and examined for conformance with the preservation, packaging, packing, labeling and marking required in the contract or order. Samples shall be selected in accordance with MIL-STD-105, Inspection level II, AQL 6.5 expressed in terms of defects per hundred unit.

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Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistant with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from recovered materials.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Packaging (DOD). The packaging shall be in accordance with ASTM Standard D3951 'Packaging, Commercial'.

Notes:

Purchasers should specify outside diameter, anchor (arbor) hole size, open or closed coat, abrasive, grit size and if certification of grading and breaking strength is acceptable in lieu of lot-by-lot testing.

The issues of ANSI B74.18, ASTM D828, ASTM D3951 and MIL-STD-105 in effect on the date of the solicitation or purchase order shall be used to determine compliance with these requirements.

ANSI standards are available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

ASTM standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

Copies of MIL-STD-105 may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

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MILITARY INTERESTS:

Military Coordinating Activity:

Navy - YD

Custodians:

Army - MR
Navy - YD
Air Force - 99

User activity:

Navy - SH

CIVIL COORDINATING ACTIVITY:
NASA - JFK

PREPARING ACTIVITY:

GSA-FSS

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